Replication of a Macroeconomic DSGE (from Herbst and Schorfheide 2016): A Bayesian Estimation under SMC in Python

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1 New Keynesian Dynamic Stochastic General Equilibrium Model

A standard small NK DSGE in Herbst / Schorfheide 2016:

$$\textbf{IS (Euler):} \quad y_t = \mathbb{E}_t y_{t+1} - \frac{1}{\tau}\Big(R_t - \mathbb{E}_t \pi_{t+1} - z_t\Big) + g_t \tag{1}$$

$$\textbf{NKPC:} \quad \pi_t = \beta\, \mathbb{E}_t \pi_{t+1} + \kappa\, y_t \tag{2}$$

$$\textbf{Taylor rule:} \quad R_t = \rho_R R_{t-1} + (1-\rho_R)\big(\psi_1 \pi_t + \psi_2 y_t\big) + \varepsilon_t^R \tag{3}$$

$$\textbf{Shocks (AR(1)):} \quad g_t = \rho_g g_{t-1} + \sigma_g \varepsilon_t^g, \qquad z_t = \rho_z z_{t-1} + \sigma_z \varepsilon_t^z \tag{4}$$

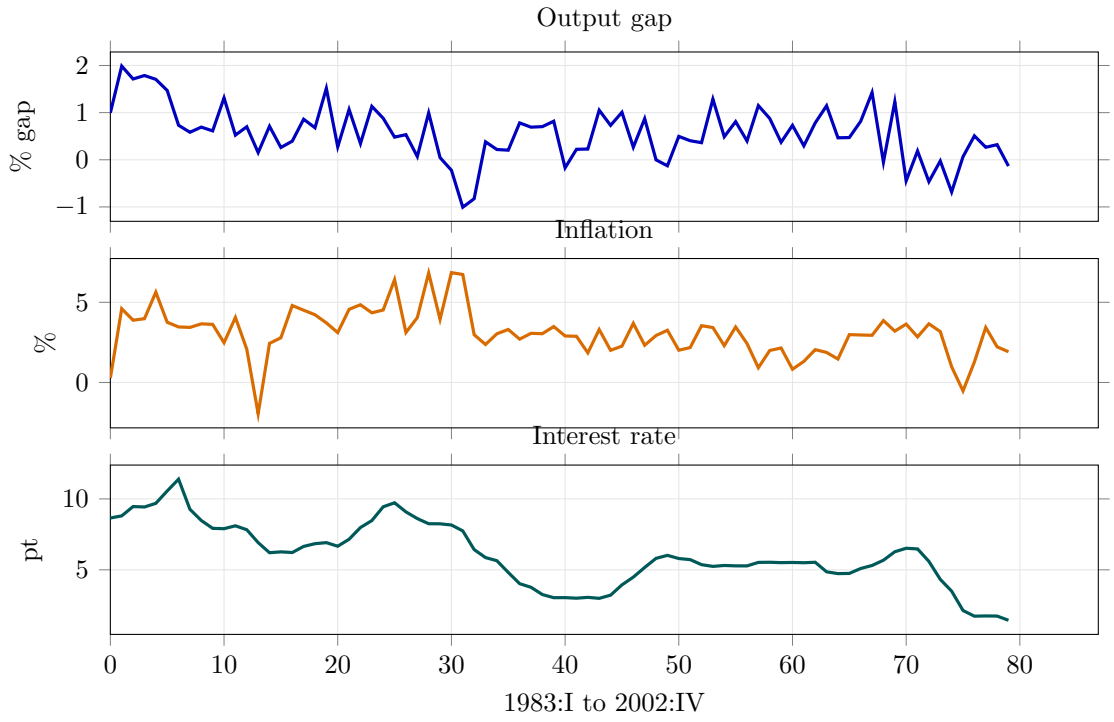


Figure 1: US data used for estimation: output gap, inflation, and nominal interest rate.

2 Goal and Inputs/Outputs

Goal

Approximate the posterior distribution of DSGE parameters θ given observed data $y_{1:T}$:

$$p(\theta \mid y_{1:T}) \propto p(\theta)\, p(y_{1:T} \mid \theta).$$

Because the likelihood evaluation is expensive (DSGE solution + Kalman filter), Herbst/Schorfheide use a *tempered Sequential Monte Carlo (SMC)* sampler with resampling and Random-Walk Metropolis–Hastings (RWMH) mutation steps.

Inputs

- **Data** $y_{1:T} \in \mathbb{R}^{T \times n_y}$ (in code: `data`)
- **Prior specification** $p(\theta)$ (in code: `log_prior`, `draw_one_prior`)
- **DSGE model code** (in code: `model_solution`, `gensys`)
- **State-space mapping and likelihood** (in code: `sysmat`, `kalman_loglik`)
- **SMC settings**: number of particles N, tempering schedule $\{\phi_t\}_{t=0}^{K}$, ESS threshold, RWMH tuning.

Outputs

- Particle approximation to posterior: $\{(\theta_i^{(K)}, w_i^{(K)})\}_{i=1}^{N}$ at $\phi_K = 1$
- Stored trajectories over tempering stages (often): particles, weights, ESS, acceptance rates
- Posterior summaries: means, credible intervals, histograms, etc.

3 Overview

Concept

1. Draw N parameter particles from the prior: $\theta_i^{(0)} \sim p(\theta)$.

```
def model_solution(theta):
    tau,kappa,psi1,psi2,rA,piA,gammaQ,rho_R,rho_g,rho_z,sigma_R,sigma_g,sigma_z =
        theta
    bet = 1.0/(1.0 + rA/400.0)
```

   - $\theta = (\tau, \kappa, \psi_1, \psi_2, r_A, \pi_A, \gamma_Q, \rho_R, \rho_g, \rho_z, \sigma_R, \sigma_g, \sigma_z)$
   - Discount factor is computed as: $\beta = \frac{1}{1+r_A/400}$

2. For each θ_i, solve DSGE $\to$ build SSM $\to$ compute $\log L_i = \log p(y_{1:T} \mid \theta_i)$.
3. For tempering stages $t = 1, \ldots, K$ with $0 = \phi_0 < \cdots < \phi_K = 1$:
   (a) **Reweight** using incremental likelihood power $\Delta\phi_t = \phi_t - \phi_{t-1}$.
   (b) Compute ESS; **resample** if ESS below threshold.
   (c) **Mutate** each particle with an RWMH kernel targeting the *tempered posterior*

$$\pi_{\phi_t}(\theta) \propto p(\theta)\, p(y_{1:T} \mid \theta)^{\phi_t}.$$

4. At $t = K$ (where $\phi_K = 1$), particles approximate the true posterior.

Tempered posterior (target at stage t)

$$\log \pi_{\phi_t}(\theta) = \log p(\theta) + \phi_t \log p(y_{1:T} \mid \theta) + \text{const}.$$

4 Evaluation from θ to output

The function `log_posterior_tempered` is used by both the SMC reweighting and the RWMH mutation steps. It outputs parameters into the triplet:

$$(\log \pi_\phi(\theta),\ \log L(\theta),\ \log p(\theta)).$$

```
def log_posterior_tempered(theta, phi):
    lp = log_prior(theta)
    if not np.isfinite(lp):
        return -np.inf, -np.inf, -np.inf

    T1, T0, GEV = model_solution(theta)
    if T1 is None or GEV[1] < 0.5:
        return -np.inf, -np.inf, lp

    A, B, H, R, Se, Phi = sysmat(T1, T0, theta)
    ll = kalman_loglik(A, B, H, R, Se, Phi, data)
    if not np.isfinite(ll):
        return -np.inf, -np.inf, lp

    return phi*ll + lp, ll, lp
```

Inputs and outputs of `log_posterior_tempered`

Symbol	Code name	Meaning
θ	`theta`	parameter vector
ϕ	`phi`	tempering exponent in $[0, 1]$
$\log p(\theta)$	`lp`	log prior
$\log p(y \mid \theta)$	`ll`	log likelihood (Kalman filter)
$\log \pi_\phi(\theta)$	`phi*ll + lp`	tempered log posterior

In all failure cases, the function returns $-\infty$ so the particle gets zero weight and MH proposals are rejected.

5 Calculation of the likelihood

5.1 DSGE solution step (linear RE model)

The DSGE is assembled and solved in a linear rational-expectations form (`gensys`):

$$\Gamma_0 x_t = \Gamma_1 x_{t-1} + C + \Psi \varepsilon_t + \Pi \eta_t,$$

```
G1_sol, C_sol, impact, *_rest, gev = gensys(GAM0, GAM1, C=C, PSI=PSI, PI=PPI, div
    =1.01)

if G1_sol is None:
    return None, None, np.array([float(gev[0]), 0.0])

return G1_sol, impact, np.array([float(gev[0]), 1.0])
```

and the solver returns a latent state transition representation:

$$x_t = \Phi x_{t-1} + R\varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, S_e)$$

where ε_t are structural shocks, with diagonal $S_e = \mathrm{diag}(\sigma_z^2, \sigma_g^2, \sigma_R^2)$.

5.2 State-space model

$$x_t = \Phi x_{t-1} + R\varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, S_e)$$
$$y_t^{obs} = A + Bx_t + u_t, \quad u_t \sim \mathcal{N}(0, H)$$

```
A = np.zeros((ny,1))
A[0,0] = gammaQ
...
A[2,0] = piA + rA + 4*gammaQ

B = np.zeros((ny, Phi.shape[0]))
B[0, y_t] = 1
B[0, y1_t] = -1
B[0, z_t] = 1
B[1, pi_t] = 4
B[2, R_t] = 4
```

So the **measurement equation** is:

$$y_t^{obs} = A + Bx_t + u_t, \qquad u_t \sim \mathcal{N}(0, H)$$

with :

$$y_{1,t}^{obs} = \gamma_Q + y_t - y1_t + z_t + u_{1,t}$$
$$y_{2,t}^{obs} = (\pi\text{-obs}) = 4\pi_t + u_{2,t}$$
$$y_{3,t}^{obs} = (R\text{-obs}) = 4R_t + u_{3,t}$$

```
H = np.zeros((ny,ny))
H[0,0] = (0.20*0.579923)**2
H[1,1] = (0.20*1.470832)**2
H[2,2] = (0.20*2.237937)**2
```

and the **state transition** is:

$$x_t = \Phi x_{t-1} + R\varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, S_e).$$

```
def sysmat(T1, T0, theta):
    ...
    Phi = T1
    R = T0 # impact on eps

    Se = np.zeros((3,3))
    Se[0,0] = sigma_z**2
    Se[1,1] = sigma_g**2
    Se[2,2] = sigma_R**2
```

In code, these are built by `sysmat(T1, T0, theta)` which returns

$$(A, B, H, R, S_e, \Phi).$$

5.3 Kalman filter likelihood

Given (A, B, H, R, S_e, Φ) and data $y_{1:T}$, the Kalman filter computes:

$$\log p(y_{1:T} \mid \theta) = \sum_{t=1}^{T} \log p(y_t \mid y_{1:t-1}, \theta).$$

via the `kalman_loglik(...)` function

6 SMC details: reweight, resample, mutate

6.1 Tempering schedule

The temperature schedule is:

$$0 = \phi_0 < \phi_1 < \cdots < \phi_K = 1,$$

with a power transform to slow down early increments ($\phi_t = (t/K)^\lambda$).

6.2 Reweighting step

Given weights $w_i^{(t-1)}$ and stored $\log L_i$, the incremental update is:

$$\widetilde{w}_i^{(t)} = w_i^{(t-1)} \exp\big((\phi_t - \phi_{t-1}) \log L_i\big), \qquad w_i^{(t)} = \frac{\widetilde{w}_i^{(t)}}{\sum_j \widetilde{w}_j^{(t)}}.$$

For here, this is why the code returns `ll` separately: reweighting can be done without recomputing the DSGE+Kalman steps.

6.3 ESS and resampling threshold

The effective sample size (ESS) is:

$$\mathrm{ESS}^{(t)} = \frac{1}{\sum_{i=1}^N \left(w_i^{(t)}\right)^2}.$$

Resampling condition

$$\mathrm{ESS}^{(t)} < \tau_{\mathrm{ESS}}\, N,$$

where $\tau_{\mathrm{ESS}} \in (0,1)$ is the ESS threshold. When resampling is true, indices are drawn (systematic resampling), particles are duplicated accordingly, and weights are reset to uniform:

$$w_i^{(t)} := \frac{1}{N}.$$

6.4 Mutation step (which is RWMH at each tempering stage)

After resampling, each particle is moved with an RWMH kernel that leaves the tempered posterior invariant.

Proposal:

$$\theta' = \theta + \epsilon, \qquad \epsilon \sim \mathcal{N}(0,\ c^2 \Sigma_t),$$

where Σ_t is the empirical covariance of the particle cloud at stage t and c is a scaling factor.

Acceptance probability: Let $\log \pi_{\phi_t}(\theta)$ be the value returned by `log_posterior_tempered(theta, phi_t)`. Then:

$$\rho(\theta, \theta') = \min\left\{1,\ \exp\Big(\log \pi_{\phi_t}(\theta') - \log \pi_{\phi_t}(\theta)\Big)\right\}.$$

Accepted proposals replace the particle; rejected proposals keep the current value.

- **Resampling** prevents weight degeneracy (a few particles carrying all mass).
- **Mutation** prevents particle impoverishment (many duplicated particles after resampling) and adapts particles to the more pronounced target as ϕ increases.

7 Inputs and Outputs

	Inputs	**Outputs**
Prior draw	Prior hyperparameters / bounds	$\theta_i^{(0)} \sim p(\theta)$
Model solution	θ	$(T1, T0, GEV)$
State-space build	$(T1, T0, \theta)$	(A, B, H, R, S_e, Φ)
Kalman likelihood	$(A, B, H, R, S_e, \Phi, \text{data})$	$\log L(\theta)$
Tempered posterior eval	(θ, ϕ)	$(\log \pi_\phi(\theta), \log L(\theta), \log p(\theta))$
SMC reweight	$\{w_i, \log L_i\}, \Delta\phi$	Updated $\{w_i\}$, ESS
Resample (conditional)	$\{w_i\}$, ESS threshold	Resampled particles, uniform weights
RWMH mutate	Particles, ϕ, proposal covariance	Moved particles, acceptance rates

8 Posterior Approximation

At the final stage $\phi_K = 1$, the particle system approximates

$$p(\theta \mid y_{1:T}).$$

Expect posterior summaries to be computed from weighted particles, e.g. posterior mean:

$$\mathbb{E}[\theta \mid y] \approx \sum_{i=1}^{N} w_i^{(K)} \theta_i^{(K)}.$$

9 Results

N = 5000 K = 400

Table 1: Posterior summaries: SMC (5000 particles, 400 tempering steps) vs. Herbst & Schorfheide (2016, Table 4.1).

	SMC		Herbst & Schorfheide (2016)	
Parameter	Mean	$[q_{0.05}, q_{0.50}, q_{0.95}]$	Mean	$[q_{0.05}, q_{0.95}]$
τ	2.3606	[1.5860, 2.3237, 3.2720]	2.83	[1.95, 3.82]
κ	0.8465	[0.6102, 0.8741, 0.9886]	0.78	[0.51, 0.98]
ψ_1	1.9211	[1.5551, 1.9146, 2.3035]	1.80	[1.43, 2.20]
ψ_2	0.5894	[0.2042, 0.5491, 1.1155]	0.63	[0.23, 1.21]
r_A	0.4417	[0.0510, 0.4101, 0.9379]	0.42	[0.04, 0.95]
π_A	3.3814	[2.7574, 3.3778, 4.0092]	3.30	[2.78, 3.80]
γ_Q	0.5988	[0.3681, 0.6018, 0.8195]	0.52	[0.28, 0.74]
ρ_R	0.7727	[0.7106, 0.7747, 0.8280]	0.77	[0.71, 0.82]
ρ_g	0.9779	[0.9456, 0.9815, 0.9984]	0.98	[0.96, 1.00]
ρ_z	0.9240	[0.8831, 0.9247, 0.9634]	0.88	[0.84, 0.92]
σ_R	0.2180	[0.1788, 0.2156, 0.2661]	0.22	[0.18, 0.26]
σ_g	0.6551	[0.5669, 0.6515, 0.7558]	0.71	[0.61, 0.84]
σ_z	0.1997	[0.1651, 0.1977, 0.2401]	0.31	[0.26, 0.36]

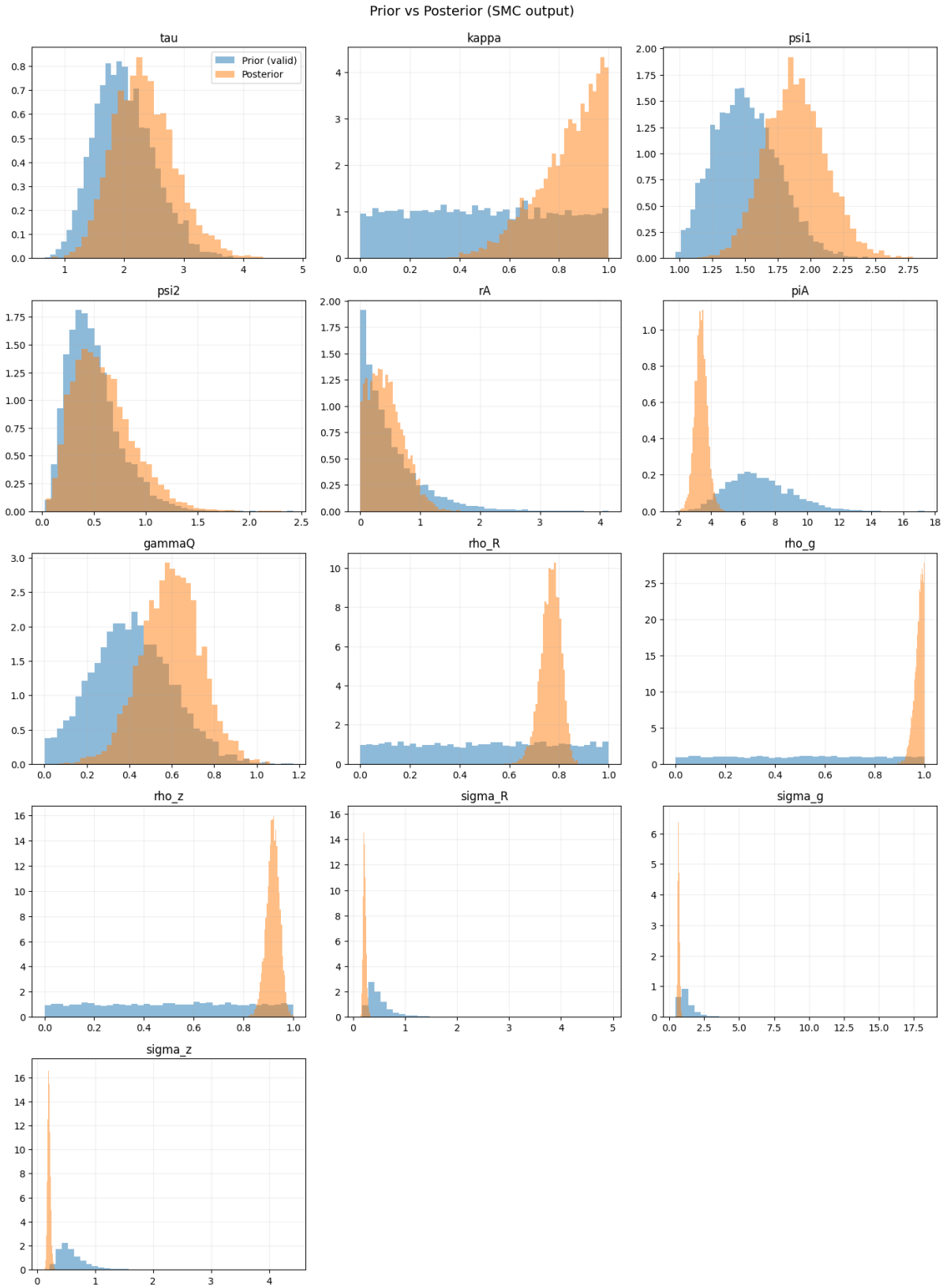


Figure 2: prior vs posterior

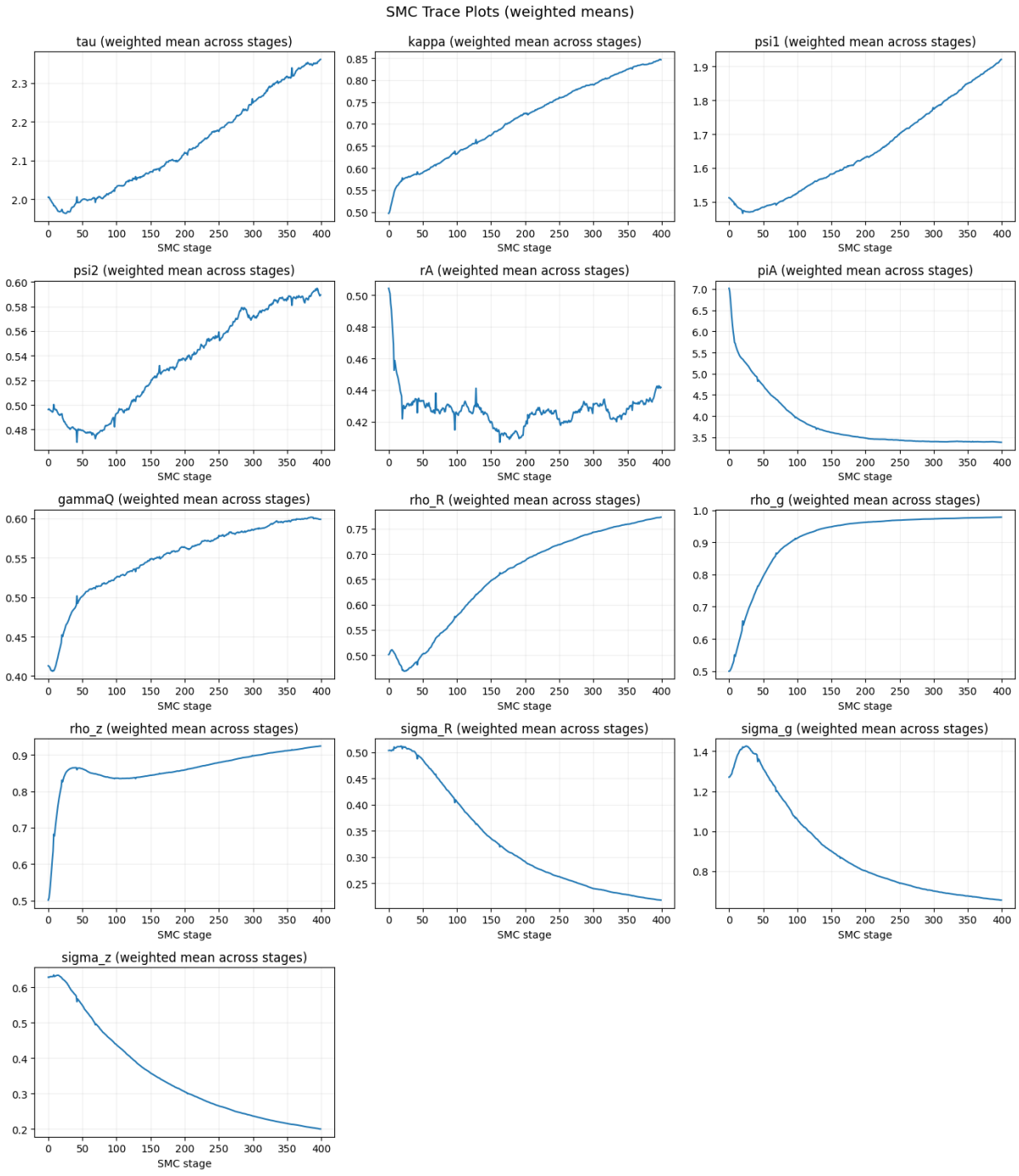


Figure 3: trace plots

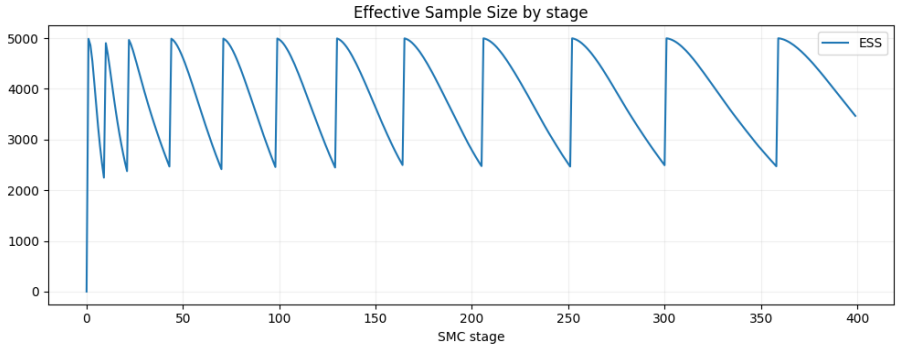


Figure 4: ESS

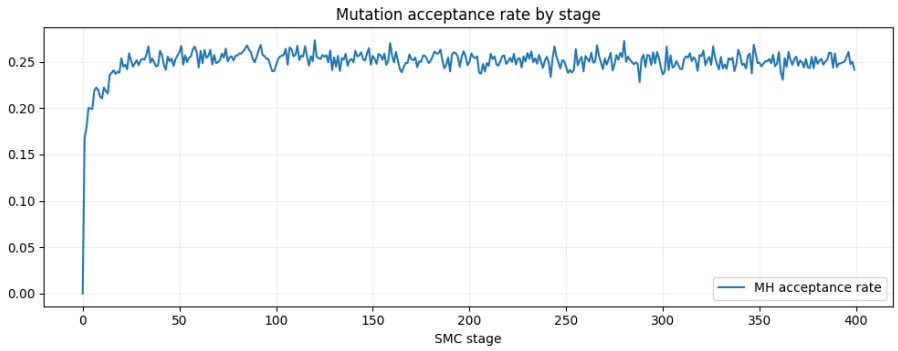


Figure 5: acceptance rate

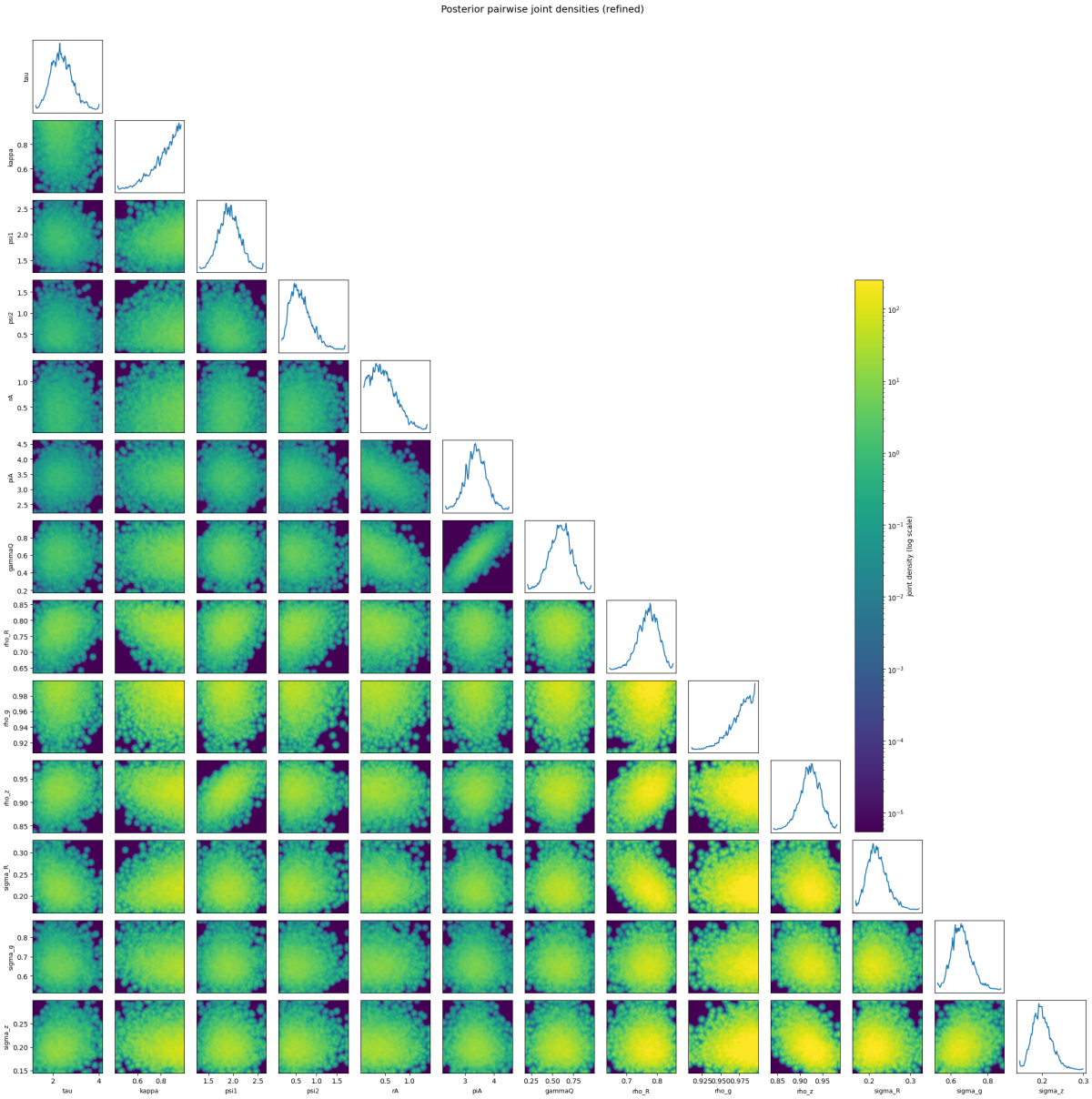


Figure 6: posterior

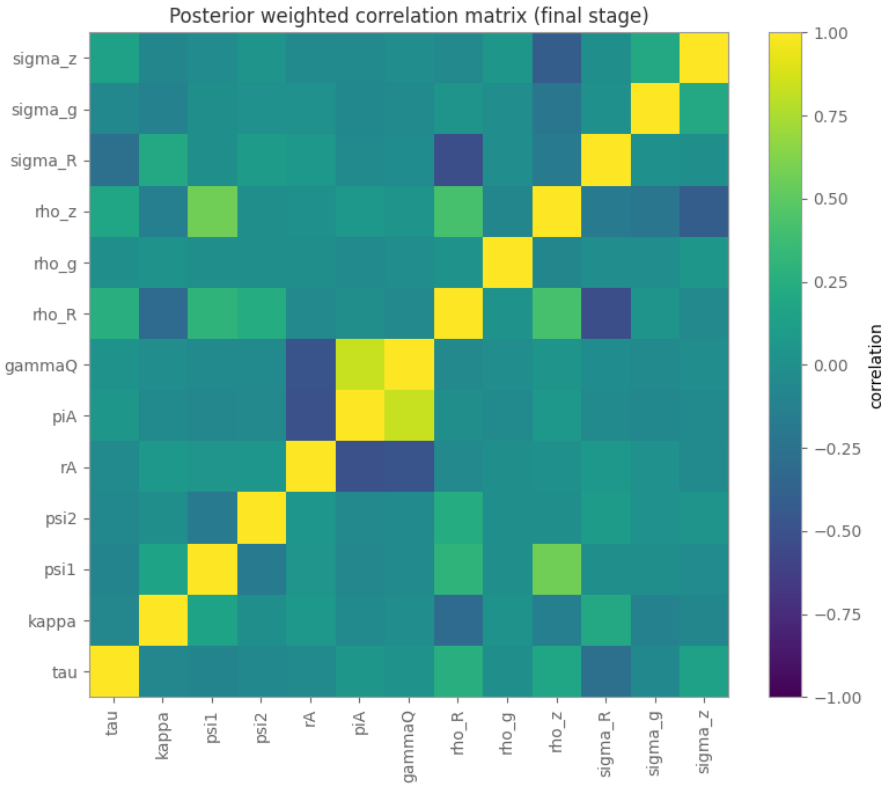


Figure 7: Correlation

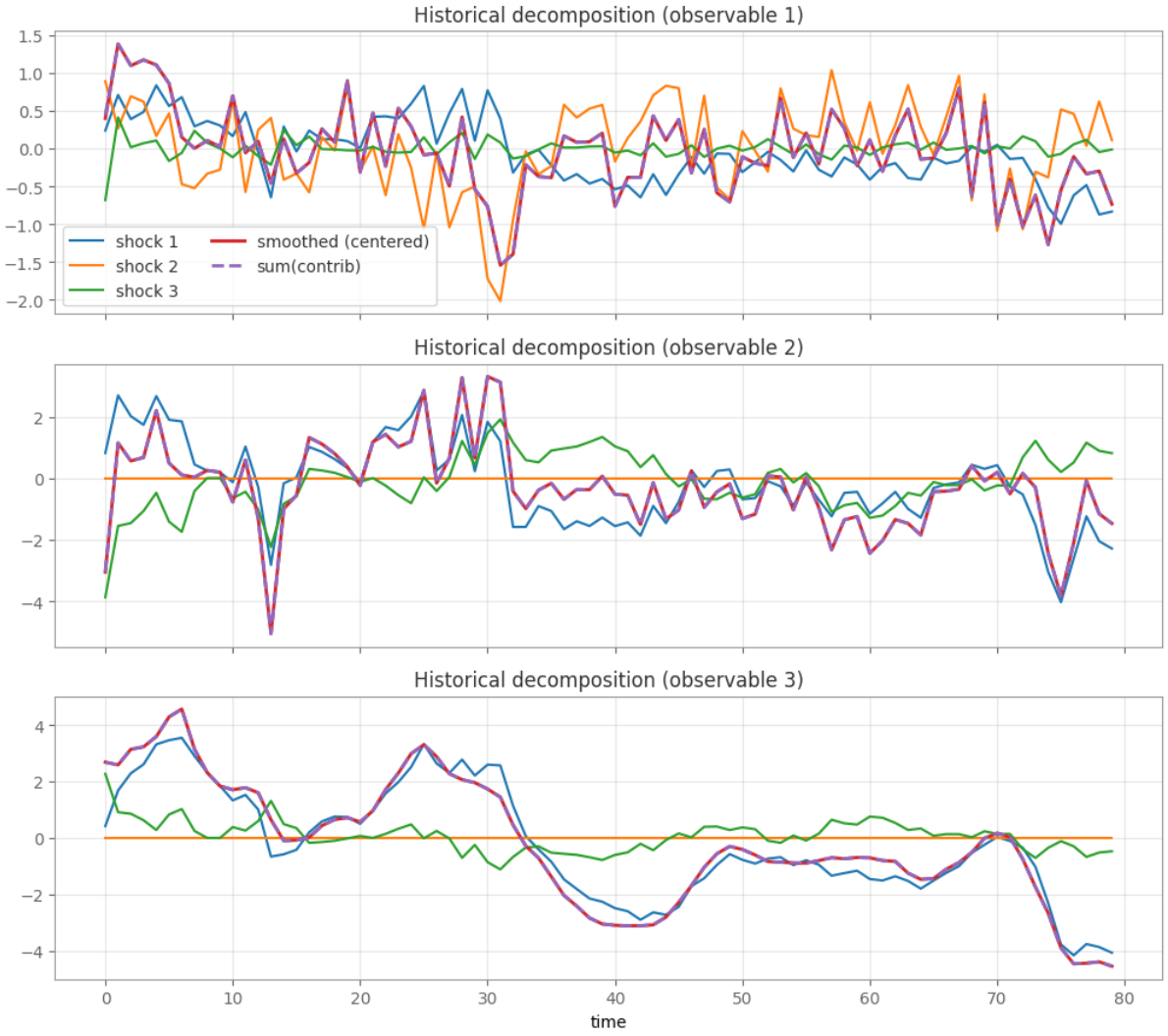


Figure 8: historical decomposition